http://https://chat.whatsapp.com/FmGdNtjc2pMEpCHnzpVjdO

Q

TH2 cells mainly help to activate :

a) T cells
b) B cells
c) MHC I molecules
d) MHC II molecules

Q
Which of the following stabilizes cell shape
a) DNA
b) Actin
c) Myosin
d) RNA

Q

Fibroblast cells can be induced to proliferate by treatment with .....
a) Platelet-derived growth factor (PDGF)
b) Epidermal growth factor (EGF)
c) Both (EGF) and (PDGF)
d) None

Q
Even in the absence of antigen humans can make more than..... different antibody molecules.

a) 10^{120}
b) 12^{10}
c) 10^{12}
d) 10^{112}

Q

The _____immune system, like the ____system, can remember prior experiences

a) Adaptive, nervous

b) Adaptive, respiratory

c) Innate, nervous

d) Activated, metabolic

Q

Inhibitor that binds to a site other than active site of enzyme

a) Competitive inhibitor
b) Reversible inhibitor
c) IRReversible inhibitor
d) Non competitive inhibitor

Q

Condition in which cells have specific binding

a) cell recognition
b) cell communication
c) cell adhesion
d) a & c

Q

In circular form, glucose molecule can adapt alpha or beta configuration. In beta glucose, OH group is pointing ------ while in case of alpha OH group is in --------- direction

a) Upward, downward
b) Downward, upward
c) Upward, linear
d) Upward, linear

Q

People with additional Y chromosome have external appearance

a) Abnormal
b) Normal
c) None
d) May be normal or abnormal

Q

Which one is the product of genetic engineering

a) Insulin
b) Quina
c) Plasmid
d) Fatty acids

Q

Which one is initiation codon

a) UGA
b) AUG
c) UAG
d) UAA

Q
Function of Golgi apparatus is

a) Packaging and transportation of proteins
b) egradation of proteins
c) Homeostasis
d) Cell movement

Q
An average globular cell have ------ mitochondria

a) 100,000
b) 10,000
c) 1,000
d) 100

Q
Chromosome fibres consist of

a) DNA and protein
b) phosphate bonds
c) nucleosomes
d) protein only

Q
Microtubules which are not attached to the chromosomes are

a) Mitotic Centre
b) Centrosomes
c) Polar microtubules
d) Polar microtubules

Q
The number of appearance of chromosomes in the nucleus of a eukaryotic cell is called

a) Prototype
b) Genetic makeup
c) Karyotype
d) Karyotype

Q
HIV is

a) DNA virus
b) RNA virus
c) Herpes virus
d) Adeno virus

Q

Structure formed by many folding of polypeptide chain

a) Primary protein structure
b) Secondary protein structure
c) Tertiary protein structure
d) Quaternary protein structure

Q
Ribosomes are engaged in:

a) Lipid synthesis
b) Protein synthesis
c) DNA synthesis
d) Sugar synthesis

Q

_____ are removed during RNA maturation.

a) Introns
b) Exons

c) Intron-Exon boundaries

d) Promoter region only

Q
TH2 cells mainly help to activate :

a) T cells
b) MHC I molecules
c) B cells
d) MHC II molecules

Q
Mendel donated dominant and recessive traits as
a) Ss and ss
b) SS and ss
c) ss and Ss
d) SS and SS

Q
Structure of protein is
a) One dimensional
b) Two dimensional
c) Three dimensional
d) Four dimensional

Q
The fusion of membranes from same compartment is termed as
a) Heterotypic fusion
b) Homotypic fusion
c) Hemitypic fusion
d) Hemitypic fusion

Q
Which one of the following substances in immune system causes capillaries to dilate and become leaky?

a) Amino acids
b) Thymine
c) Carbohydrates
d) Histamine

Q
On which site a substrate binds on an enzyme.....
a) Peptidal site
b) Active site

c) Binding site
d) Catalytic site

Q
For optimal activity of hydrolytic enzymes, lysosome maintains a pH about
a) 7.0
b) 5.0
c) 3.0
d) 9.0

Q
Which one of the following helps in stimulating breast cells to produce milk?
a) STAT 1
b) Erythropoietin
c) Prolactin
d) Interferon

Q
Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it
a) Transmembrane proteins
b) Water soluble proteins
c) Extrinsic proteins
d) Extrinsic proteins

Q
Activation energy of an enzymatic reaction is
a) Low
b) High
c) Intermediate
d) All

Q
Ran GTPase is a molecular switch that exist in
a) 2 states
b) 6 states
c) 8 states
d) 4 states

Q
----------is one of the example of Coenzyme
a) Iron
b) Zinc
c) NAD
d) Haeme

Q
An oncogene after mutation produces one of abnormal condition

a) Kidney stones
b) Encephalopathy
c) Diabetes
d) Cancer

Q
Enzyme used in PCR process

a) *Thermus hydrophila*
b) *Thermus punctata*
c) *Thermus aquaticus*
d) *Thermus fario*

Q
Bacterial DNA is present in
a) Nucleoid
b) Nucleolus
c) Nucleus
d) Nucleoplasm

Q
Dendritic cells use on of following signal to guide helper T cells to effector T cells
a) IL-4
b) IL-3
c) IL-2
d) IL-1

Q
Genes are located in the nucleus and tangled charamatin material wrapped around histones in .....
a) Both Eukarotes and Prokaryotes
b) Eukaryotes
c) Prokaryotes
d) None of these

Q
Organisms shuffle genes by
a) Meiosis
b) Mitosis
c) binary fission
d) multiple fission

Q
Inactive "Intracellular Receptors" (ICR) may occur .....

a) free moving in plasma
b) Occur in cytoplasm
c) Occur as DNA bound
d) Both (Occur in cytoplasm) and (DNA bound)

Q
The cells which are activated initially by recognizing foreign antigen on APC
a) B cells
b) Macrophages
c) B and T cells
d) T cells

Q
a) Lipids are……..
a).... insoluble, hydrophobic molecule
b) soluble, hydrophilic molecules
c) basic molecule
d) acidic molecule

Q
How many times an average cell divide before it gets unhealthy
a) 25 times
b) 50 times
c) 75 times
d) 100 times

Q

_____ are removed during RNA maturation.
a) Introns
b) Promoter region only
c) Intron-Exon boundaries
d) Exons

Q
T.H.Morgan observed

a) Independent assortment of genes
b) Gene linkage
c) Segregation of genes
d) Fertilization

Q
Cholera toxin prevents.....

a) Hydrolysis of GTP to GDP
b) Formation of ATP

c) Formation of GTP
d) None of these

Q
Which one is larger sex chromosome
a) X
b) Y
c) XY
d) None

Q
Stitching two different DNAs together is
a) Replication
b) Genetic engineering
c) Translation
d) Transcription

Q
Philadelphia chromosome is a chromosomal translocation between chromosome ___ and _____.
a) 5 & 6
b) 7 & 21
c) 1 & 2
d) 9 & 22

Q
Cyanobacteria are type of bacteria that have
a) Cyanocide
b) Nucleus
c) Chlorophyll
d) Mitochondria

Q
The _____immune system, like the ____system, can remember prior experiences
a) Adaptive, respiratory
b) Adaptive, nervous
c) Innate, nervous
d) Activated, metabolic

Q
Chromosome fibres consist of
a) protein only
b) nucleosomes
c) DNA and protein
d) phosphate bond

Q
How many types of bacterial colony were used by Freddrick Griffith in his experiment

a) 2
b) 4
c) 6
d) 8

Q
Endoplasmic reticulum is composed of rough and smooth endoplasmic reticulum and plays important role in ----------

a) Sorting and modifying protein
b) Sorting protein
c) Modifying protein
d) Degrading protein

Q
There are ..... lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain

a) 2×12^{10}
b) 2×10^{21}
c) 2×10^{12}
d) 12×10^{12}

Q
Genes are located in the nucleus and tangled charamatin material wrapped around histones in .....

a) Both Eukarotes and Prokaryotes
b) Eukaryotes
c) Prokaryotes
d) None of these

Q
If one gene prevent the expression of other gene

a) Pleiotropy
b) Epistasis
c) Synapse
d) Tetrad

Q
The fusion of membranes from same compartment is termed as

a) Heterotypic fusion

b) Homotypic fusion
c) Hemitypic fusion
d) Hypotypic fusion

Q
Approximately, How many years ago life orignated.....
a) 4,000,000,000
b) 4,000,000,000, 000,000
c) 4,000,000,000,000
40,000,000,000

Q
Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it
a) Intinsic proteins
b) Extrinsic proteins
c) Water soluble proteins
d) Water soluble proteins

Q
Phenylalanine hydroxylase metabolizes
a) Phenylalanine to Lysine
b) Phenylalanine to Tyrosine
c) Tyrosine to Phenylalanine
d) Lysine to Phenylalanine

Q
Activation of Phospholipase C-β
a) 10%
b) 90%
c) 20%
d) 1%

Q
Mental retardation is caused by
a) Leishmaniasis
b) Cystic fibrosis
c) Phenylketonuria
d) Sickle cell anemia

Q
The agent that is used as cryoprotectant is

a) KMNO4
b) Acetic acid
c) EDTA
d) DMSO

Q:
cell membrane intake particles by vesicle formation
a) Endocytosis
b) Exocytosis
c) Pinocytosis
d) Phagocytosis

Q
In 2014 it was reported that one of following proteins in blood is helpful in cure of Alzheimer's disease and heart disease
a) A 11
b) A12
c) A13
d) A14

Q
Undefferentiated cells of multicellular organisms which can develope into specialized cells
a) Hepatocytes
b) Degenerative cells
c) Mast cells
d) Stem cells

Q
Autocrine signals bind to the receptors on the cells, which .....
a) are arround them
b) has no involvement of the receptors
c) Adjacent Cells
d) Cells that secrete them

Q
People with additional Y chromosome have external appearance
a) Normal
b) Abnormal
c) May be normal or abnormal
d) None

Q
Helices that are/is involved in agonist binding are
a) Helix 6
b) Helix 5
c) Helix 3
d) All

Q
The lipids molecules held together by weak interactions and
a) Van der waals forces
b) Hydrogen bonds
c) Isotonic bonds
d) Esterase bonds

Q
β-blockers (antagonist) are involved in
a) Bind to receptor by not activating it
b) None of these
c) Mimicking the function of a hormone
d) Making the hormone activity better

Q
a) Initiation site of transcription
b) Co factor
c) Promoter
d) Active site
e) Effective site

Q
Sex related disorders are
a) X- related
b) Y- related
c) XY- related
d) XYZ - related

Q
Which one is initiation codon
a) UAG
b) AUG
c) UGA
d) UAA

Q
Function of G protein in vision may involve.....

a) Removal of NADPH
b) Production of ATP
c) Inactivation of K+ channels
d) Activation of Phospholipase C-β

Q
Calmodulin is about.....% of total cell protein mass
a) 10%
b) 20%
c) 90%
d) 1%

Q
The agent that is used as cryoprotectant is
a) Acetic acid
b) KMNO4
c) DMSO
d) EDTA

Q
In prometaphase I, only ONE of the following occurs:
a) Sister chromatids start pairing up
b) Centrosomes start moving to the opposite poles and form microtubules
c) Condensation of DNA material
d) Disintegration of the nuclear membrane

Q
In 2014 it was reported that one of following proteins in blood is helpful in cure of Alzheimer's disease and heart disease
a) A11
b) A12
c) A 13
d) A 14

Q
Undefferentiated cells of multicellular organisms which can develope into specialized cells
a) Mast cells
b) Hepatocytes
c) Stem cells
d) Degenerative cells

Q
Once the genetic information has passed on to the next generation, it does not
a) change

b) inherit to next generation
c) remove
d) remove

Q
Both (Occur in cytoplasm) and (DNA bound)
a) Addition of phosphate group
b) Removal of phosphate group
c) Both addition and removal and thrionine residues of phosphate group
d) Phosphorylation of specific serine

Q
The fusion of membranes from same compartment is termed as
a) Heterotypic fusion
b) Hemitypic fusion
c) Homotypic fusion
d) Hypotypic fusion

Q
For optimal activity of hydrolytic enzymes, lysosome maintains a pH about
a) 3.0
b) 3.0
c) 5.0
d) 7.0

Q
Phosphorylation signaling in the E. coli alters
a) Kinase acctivity
b) Transcription factor
c) Responder function
d) Protein synthesis

Q
Proteins which are partly translocated across Endoplasmic reticulum and are embedded in it
a) Transmembrane proteins
b) Water soluble proteins
c) Intrinsic proteins
d) Extrinsic proteins

Q
Ran GTPase is a molecular switch that exist in
a) 2 states
b) 4 states
c) 6 states
d) 8 states

Q
How many protein molecules an animal cell contains
a) 5 billion
b) 10 billion
c) 15 billion
d) 20 billion

Q
Autocrine signals bind to the receptors on the cells, which .....
a) are arround them
b) Cells that secrete them
c) Adjacent Cells
d) has no involvement of the receptors

Q
An oncogene after mutation produces one of abnormal condition
a) Diabetes
b) Kidney stones
c) Encephalopathy
d) Cancer

Q Dendritic
cells use on of following signal to guide helper T cells to effector T cells
a) IL-1
b) IL-3
c) IL-4
d) IL-2

Q
The cells which are activated initially by recognizing foreign antigen on APC
a) B cells
b) B and T cells
c) Macrophages
d) T cells

Q
_____are passed on to the daughter cells after mitosis but not passed on to the children.
a) Germ line mutations
b) Point mutations
c) Somatic mutations
d) Chromosomal mutations

Q
Philadelphia chromosome is a chromosomal translocation between chromosome ___ and _____.
a) 7 & 21
b) 1 & 2
c) 9 & 22

d) 5 & 6

Q
The _____immune system, like the ____system, can remember prior experiences

a) Adaptive, nervous
b) Adaptive, respiratory
c) Innate, nervous
d) Activated, metabolic

Q
Even in the absence of antigen humans can make more than..... different antibody molecules.

a) 10^{120}
b) 12^{10}
c) 10^{12}
d) 10^{112}

Q
Undefferentiated cells of multicellular organisms which can develope into specialized cells

a) Mast cells
b) Stem cells
c) Hepatocytes
d) Degenerative cells

Q
For optimal activity of hydrolytic enzymes, lysosome maintains a pH about

a) 3.0
b) 5.0
c) 7.0
d) 9.0

Q
Ran GTPase is a molecular switch that exist in

a) 2 states
b) 4 states
c) 6 states
d) 8 states

Q
How many protein molecules an animal cell contains

a) 5 billion
b) 10 billion
c) 15 billion
d) 20 billion

Q
There are ..... lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain

a) 2×12^{10}
b) 2×10^{12}
c) 2×10^{21}
d) 12×10^{12}

Q
An oncogene after mutation produces one of abnormal condition
Diabetes

a) Kidney stones
b) Encephalopathy
c) Encephalopathy
d) Cancer

Q
The cells which are activated initially by recognizing foreign antigen on APC

a) B cells
b) T cells
c) B and T cells
d) Macrophages

Q
_____are passed on to the daughter cells after mitosis but not passed on to the children.

a) Germ line mutations
b) Somatic mutations
c) Point mutations
d) Chromosomal mutations

Q

Philadelphia chromosome is a chromosomal translocation between chromosome ___ and _____.

a) 9 & 22

b) 1 & 2

c) 7 & 21

d) 5 & 6

Q
Which one of the following gene is mutated in approximately 30% of all human tumors?

a) BRCA1 gene
b) EFGR gene
c) RAS gene
d) all of the above

Q
Genes are located in the nucleus and tangled charamatin material wrapped around histones in .....
a) Both Eukarotes and Prokaryotes
b) Eukaryotes
c) Prokaryotes
d) None of these

Q
Even in the absence of antigen humans can make more than..... different antibody molecules.
a) 10^{120}
b) 10^{12}
c) 12^{10}
d) 10^{112}

Q
Which one of the following helps in stimulating breast cells to produce milk?

a) STAT 1
b) Interferon
c) Erythropoietin
d) Prolactin

Q
Which one of the followings protein is activated when DNA is damaged,
a) Histone
b) Kinase
c) Actin

d) Albumin

Q
Which one of the following substances in immune system causes capillaries to dilate and become leaky?

a) Amino acids
b) Thymine
c) Histamine
d) Carbohydrates

Q
Some of the genes those who remain expressed all the time in cells are called?
a) Differentially expressed genes
b) Constitutively expressed genes
c) Housekeeping genes
d) Both housekeeping and constitutive genes

Q
a) Which one of the following substances in immune system causes capillaries to dilate and become leaky?
b) Thymine
c) Amino acids
d) Histamine
e) Carbohydrates

Q
Which one of the following helps in stimulating breast cells to produce milk?
a) Erythropoietin
b) Prolactin
c) Interferon
d) STAT 1

Q
Which one of the followings protein is activated when DNA is damaged,
a) Actin
b) Kinase
c) Histone
d) Albumin

Q

What is the role of MHC in immunity? Enlist some major differences between MHC I and MHC II molecules.

Q
What is the homotypic and heterotypic fusion and how it forms?

Q
Differentiate Euchromatin and Heterochramatin

Q
What is the homotypic and heterotypic fusion and how it forms?

Q
Give the detail of events taking place at synapse at an activated neuromuscular junction?

Q
Differentiate between B cells and T cells?

Q
Please enlist the strategies of transcription regulation.....

Q
How UV light help to sterilization the equipment.....

Q

Please elaborate the different stages of the cell cycle.....

Q
How a sheep can be cloned through nuclear transfer, provide details in the form of steps?

Q
Describe the two signaling function of N-terminal ER signal sequence of a soluble protein.....

Q
How do terbutaline and β-blockers affect the function of Epinephrin adrenaline?

Q

Name five major growth promoting oncogenes.

Q
Whar are Stress Response Elements.....
.

Q
Please enlist the strategies of transcription regulation.....

Q
Give the detail of events taking place at synapse at an activated neuromuscular junction?

Q
Name five major growth promoting oncogenes.

Q
What are the Stem cells and how they differ from the specialized cells?

Q
Differentiate the "confluent monolayer" and "past confluency" stages.....

Q
Please elaborate the different stages of the cell cycle.....

Q
Please elaborate the different stages of the cell cycle.....

Q
Write detailed mechanism of TGF- β receptor family?

Q

How G protein is involved in the formation of GDP ultimately

Q

Write detailed mechanism of TGF- β receptor family?